



Ariel Gjaci

Postdoctoral Researcher in Artificial Intelligence

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Professional Links -

- Linkedin Profile
- Github Projects
- Google Scholar Profile
- Personal Website

Languages

- Italian
- Albanian
- English

About Me

Postdoctoral Researcher at the Italian Institute of Technology (IIT) with a PhD in AI and Robotics, an MSc in Robotics Engineering, and a BSc in Biomedical Engineering. Currently working on the Xtreme European project, developing multimodal ML models for realistic binaural audio rendering at any position in a room, to be integrated into mixed reality systems for immersive audio-visual experiences. My previous research focused on culture-aware co-speech gesture generation for social robots. Passionate about learning and applying AI to complex, socially impactful challenges across machine learning and robotics.

Professional Experience

- 02/2025 – now** **Postdoctoral Researcher** Italian Institute of Technology (IIT)
 - Working on audio rendering tasks in the European project XTREME.
 - Writing a survey on learning-based methods for novel-view audio rendering.
 - Designing a high-quality binaural audio renderer driven by acoustics and visual features.
 - Contributing to the release of a production-level system.
- 05/2021 – 10/2021** **Software Engineer** Akka Technologies
 - Developed backend software in Python for managing aircraft panels as part of a team project.
 - Delivered and presented a working demo to stakeholders.

Study

- 11/2021 – now** **PhD student in AI applied to robotics** Università degli Studi di Genova - King's College London
 - Built a large-scale multimodal dataset (74+ hours, 700+ subjects) with 3D pose, audio, and text across four cultures.
 - Analyzed cultural influences using machine learning models across diverse datasets.
 - Learned subject-independent cultural embeddings from multimodal features via domain generalization.
 - Developed algorithms based on semantic similarity scores from LLM cross-encoders to identify gestures in text, achieving >9% mAP and >10% IoU improvements over a statistical baseline.
 - Designed a transformer-diffusion model for autoregressive gesture generation; integrated cultural encodings to improve cultural alignment and beat sync by 5%.
- 2018 – 2021** **Master's Degree in Robotics Engineering** Università degli Studi di Genova
 - Studied subjects including computer vision, ML, control theory, embedded systems, navigation, manipulators, and ROS.
 - Graduated defending the thesis: Culture-Aware Co-Speech Gestures Using Generative Adversarial Networks.
- 2014 – 2018** **Bachelor Degree in Biomedical Engineering** Università degli Studi di Genova
 - Studied mathematics, physics, signal processing, electronics, programming (C/C++/Matlab), materials science, and physiology.
 - Graduated defending the thesis: Characterization of the Activity Evoked by Stimuli in Hippocampal Neuronal Networks.

Technical Skills

- | Programming | System Level |
|--|---|
| <ul style="list-style-type: none">PythonC++/CRMatlab/Simulink | <ul style="list-style-type: none">LinuxROSWindows |
| Machine Learning Frameworks | Embedded and Electronics |
| <ul style="list-style-type: none">PyTorchScikit-learnTensorflow | <ul style="list-style-type: none">dsPICDEM 2Raspberry PiArduino |

Soft Skills

- Learning ● ● ● ● ●
- Motivation ● ● ● ● ●
- Problem-Solving ● ● ● ● ●
- Teamwork ● ● ● ● ●
- Communication ● ● ● ● ●
- Leadership ● ● ● ● ●
- Time Management ● ● ● ● ●

Extra Activities

- Continuous learning in AI, Robotics, Programming, and related fields
- Co-organizer of a workshop at ECCV
- Reviewer for papers in AI and Robotics
- Chair of a session at the 2024 IEEE RO-MAN conference
- Presented at I-RIM 2022
- Presented at ICRA@40 event

Research Interest

Machine Learning, Robotics, Computer Vision, Computational Acoustics, Multimodal Learning, Generative Modeling, Domain Generalization, Natural Language Processing, Motion Generation, Fairness, and Reinforcement Learning.

Publications

2025	SICAGE: Subject-Independent Culture-Aware Gesture Generation with TED Four Cultures Large Dataset <i>Ariel Gjaci, Antonio Sgorbissa, Pietro Morerio, Alessio Del Bue</i> Submitted to ACM Multimedia (ACMMM)
2024	Exploring Cultural Cues in Multimodal Interactions <i>Ariel Gjaci, Viktor Schmuck, Antonio Sgorbissa, Oya Celiktutan</i> The 3rd Workshop on Affective Human-Robot Interaction (AHRI) at International Conference on Affective Computing (ACII)
2024	Labeling Sentences with Symbolic and Deictic Gestures via Semantic Similarity <i>Gjaci Ariel, Recchiuto Carmine Tommaso, and Sgorbissa Antonio</i> 33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)
2022	Towards culture-aware co-speech gestures for social robots <i>Gjaci Ariel, Recchiuto Carmine Tommaso, and Sgorbissa Antonio</i> International Journal of Social Robotics
2022	Culture Awareness in Intelligent Systems <i>Gjaci Ariel, Oneto Luca, Recchiuto Carmine Tommaso, and Sgorbissa Antonio</i> Workshop on Artificial Intelligence and Robotics – AIRO 2022
2021	A GAN-based Approach for Generating Culture-Aware Co-Speech Gestures <i>Gjaci Ariel, Recchiuto Carmine Tommaso, and Sgorbissa Antonio</i> Workshop on Artificial Intelligence and Robotics – AIRO 2021

Teaching Activity

09/2022-09/2024	Foundations of Informatics Provided hands-on assistance and tutoring on C++ basics (60 hours).	Università degli Studi di Genova
03/2024-09/2024	Mobile programming Provided hands-on assistance and tutoring in Android development using Kotlin (30 hours).	Università degli Studi di Genova

Certifications

2023	1st Doctoral Summer School on Robotics and Intelligent Machines <i>Scuola Superiore Sant'Anna</i> • Learned and revised different robotics-related topics, i.e., ROS programming, sensor data acquisition with LabView, CAD design with PTC Creo, and Computer Vision algorithms. • Worked on a team project involving the control of a 6-DOF manipulator to move a dice in the right position.
2023	Topics in Modern Machine Learning (ModML) <i>MaLGA</i> • Attended lectures on modern Machine Learning topics: Statistical Learning, Optimization, Sketching, Implicit Regularization, Reinforcement Learning, Machine Learning for Inverse Problems, Optimal Transport, Fairness, Learning in interpolation regimes, and Sampling as first-order optimization. • Completed the exam involving the completion of 3 Colab Notebooks on 3 different topics.
2022	Natural Language Processing with Classification and Vector Spaces <i>Coursera</i> • Learned basic concepts of Natural Language Processing.
2021	Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization <i>Coursera</i> • Learned basic optimization concepts of Neural Networks.